

Ground Penetrating Radars (GPR)

GPR

What is a GPR

Ground-penetrating radar (GPR) is a geophysical technique that uses radar pulses to image the subsurface.

How GPR Works

GPR works by sending a radio signal or series of signals into the ground. The returning signals provide information about changing ground characteristics with depth. Basic radar output presents this as patterns of signals at depth against the distance travelled along the ground by the radar. The radar measures depth in terms of the time it takes for a signal to return after emission. This makes GPR a very good detector of pipes & services (intrusive material), buried building materials (also intrusive) and changes in stratigraphy whether man-made (e.g. pavement profiling) or natural (e.g. peat basin detection). GPR can also be used to look inside structures either to check on the state of construction or to locate hidden objects (e.g. voids).

Why GroundVue ?

Groundvue range is designed by radar experts. They are the result of 30 years experience in advanced radar design – not only in Ground Penetrating Radar but also with avionics, military and SAR applications. Groundvue radars give you clear data quickly. They cut down on processing time: hardware design handles most of the clutter. Don't take our word for it – ask to see sample data.

Efficiency, Versatility and Toughness

These are the fastest GPR systems on the planet, transmitting simultaneously on all channels, recording positional data (GPS or Total Station) at the same time as the GPR data, allowing for fast efficient data collection. They are built to deal with a variety of environments.

Data Clarity

Clarity really matters. It's what allows you to take decisions on-site. It can make the difference between detection and non-detection, particularly in difficult environments. It can also save you time in post-processing.



User Friendliness

A simple user interface which is easy to operate. Suited to operation by individuals whose expertise and technical abilities are drawn from their own disciplines - not necessarily GPR.

Data is always recorded unprocessed. Real time viewing of the data can be in raw or processed format, but data recorded in raw format minimises the risk of inappropriately recorded data (or, worse, the need to redo a survey).

Regulation Compliance

Groundvue radars are designed to meet the ETSI specifications, EN-302 489 32. Can also meet FCC regulations if required.

Award Winning Technology

Last, but not the least, as well as the Groundvue GPRs, Utsi Electronics Ltd carry out a number of specialist designs for clients. Most of the Groundvue range of GPRs originated in this sort of development work.



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The GroundVue Range

Utsi Electronics offers, what can be considered as one of the widest range of GPRs, from extremely shallow depth scanning of few centimeters to hundreds of meters.

Table:

FREQUENCY	DEPTH	CHANNEL	CONDITIONS	RADAR
6GHz	0 to 400mm	Single	Dry	Groundvue 5C
4GHz	0 to 0.5m	Single	Dry	Groundvue 5
1.5GHz	0 to 1m	Up to 4	Dry	Groundvue 3
1.5GHz	0 to 1m	Single	Inside ducting	Borevue
1GHz	0 to 1.5m	Single,up to 4	Dry	Groundvue 3
200 to 600MHz	0 to 5m	Single,up to 4	Dry	Groundvue 3
200 to 600MHz	0 to 3m	Single,up to 4	Wetland	Groundvue 3
125 to 500MHz	0 to 10m	Single,up to 4	Dry	Groundvue 3
125 to 500MHz	0 to 5m	Single,up to 4	Wetland	Groundvue 3
30 to 100MHz	0.5 to 20m	Single	Dry	Groundvue 2
30 to 100MHz	0.5 to 11m	Single	Wetland	Groundvue 2
30 to 100MHz	0.5 to 11m	Single	Freshwater	Watervue
10 to 30MHz	0 to 200m	Single	Any	Groundvue 6
40MHz	0 to 100m	Single	Any	Groundvue 7

Groundvue 2

Groundvue 2 is a user friendly and uses easy to operate dedicated software. Operating depth (measured in travel time) and the distance between samples must be defined accurately. All other parameters can be reset after survey.

Typical Applications

For environmental monitoring including wetland & peat, and archaeological detection in wetlands.

Depth Range

At 640ns:

0.5m to 30m in dry conditions

0.5m to 11m in waterlogging

At 1280ns:

0.5m to 60m in dry conditions

0.5m to 20m in waterlogging

Resolution : 200mm

Special Features

Arrayed antennas, fully screened, suitable for towing behind a vehicle during survey. Optional GPS/Total Station data input. Data storage direct to solid disk



Ground Penetrating Radars (GPR)

GroundVue 3_8 Channel

This is the latest high speed flexible Groundvue multi-channel. It can be used in any configuration, from a trolley, from a vehicle, hand towed. Antennas can be attached, detached for another application, changed around at will. And the viewing parameters can be changed during operation. This is the radar used to detect fibre optic cables (video coming soon) and also to accurately track the direction of utilities at one sweep.

Typical Applications

The wide range of antenna frequency makes this an extremely versatile equipment. The applications range from scanning for structural integrity of RCC structures, rebar mapping, fire damage, road pavement inspection, utility mapping, archeological surveys etc. to name a few.

Antenna Frequencies

250MHz, 400MHz, 1GHz, 1.5GHz, 4GHz, 6GHz. 100MHz coming soon.

High Speed Multi-Channel

- Up to 1,600 scans per second, data collection speed consistent with traffic speeds
- Simultaneous triggering without cross channel interference
- Up to 8 channels: select any number from 1 to 8



Wi-Fi

- Wi-Fi operation, Ethernet cable for frequencies > 1GHz where Wi-Fi cannot be used

Crack Depth Measurement

Any GPR can detect areas of generalised cracking but the special crack detection head is used for:

- Accurate detection of joints in concrete
- Depth measurement of surface (top down) cracking
- Detection of subsurface (bottom up) cracking on roads, airports and in structures such as concrete loading bays and flat roofs

Automatic Velocity Calibration

- Groundvue antennas are simultaneously triggered, not sequentially, and without cross channel interference. Using multiple receivers connected to a single transmitter, continuous comparative data is created. Layer tracing generates the information needed to calculate transmission velocity for the full length of the data collected. In the process, any anomalous areas show clearly.



Groundvue 3_1

This is the single channel version of the wireless Groundvue 3_8 and can be used with any of the Groundvue 3 antennas (one at a time).

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Groundvue 5

Groundvue 5 is user friendly and uses easy to operate dedicated software. Frequency ranges are 2GHz to 6GHz .

Special Features:

- Hand held (dimensions excluding connectors and handles: 240x16x90mm)
- Lightweight
- Horn antennas
- Fully screened
- Automatic Mapping Programme available (at no extra cost)

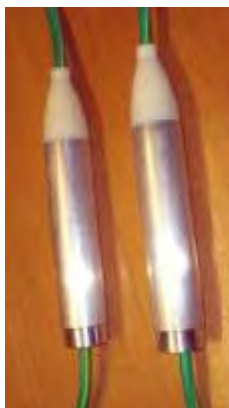
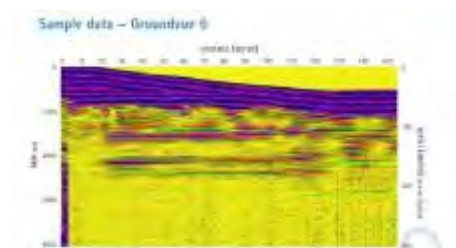


Groundvue 5C

Frequency Range is 6 GHz central (1GHz to 8GHz) and Depth Range Up to 400mm.

Special Features:

- Highest available GPR resolution capability
- Handheld
- Lightweight
- Dimensions excluding connectors and handles: 240x16x90mm
- Horn antennas.



Groundvue 6

Groundvue 6 is a thin inline spread spectrum GPR pulled behind the operator for harsh environments. Frequency Range is 7MHz to 30MHz & Depth Range is Up to 180m.

Special Features:

- Hermetically sealed
- Internal batteries
- Wireless synchronization and data transfer
- Ultrasensitive correlation receiver
- Internal data stacking (to improve signal to noise ratio)
- GPS interface

Groundvue 6C

Frequency Range 15MHz to 60MHz and Depth Range Up to 90m.

Special Features:

- A thin inline spread spectrum GPR pulled behind the operator for harsh environments
- Hermetically sealed
- Internal batteries
- Wireless synchronization and data transfer
- Ultrasensitive correlation receiver
- Internal data stacking (to improve signal to noise ratio)
- GPS interface



Groundvue 7

This low frequency (30MHz) GPR with snake antennas was formerly subject to an exclusivity agreement. As this agreement has now been cancelled, this radar is now available for hire or purchase.

